D.A.V. PUBLIC SCHOOL, SECTOR IV, B. S. CITY

CLASS XII — MATHEMATICS REVISION SHEET

TOPIC-WISE REVISION (SET A + SET B COMBINED) **TOPIC-WISE QUESTION BANK**

GENERAL INSTRUCTIONS:

1. This sheet contains questions organized strictly topic-wise from Mock Test IV (Sets A & B) to facilitate structured preparation and exam revision.
2. Work through all variations in each topic to master vector operations, 3D geometry equations, linear programming solutions, and conditional probability / Bayes' theorem applications.

UNIT 1: VECTOR ALGEBRA

Q1. If the magnitude of a non-zero vector a is p and λ is a non-zero scalar, then λa is a unit vector if:

(a) $\lambda = 1$ (b) $\lambda = -1$

(c) $p = |\lambda|$ (d) $p = 1 / |\lambda|$

Q2. If the vectors $5\mathbf{i} - \lambda\mathbf{j} + 2\mathbf{k}$ and $2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$ are perpendicular to each other, then find the value of λ:

(a) 3 (b) 4

(c) −6 (d) 0

Q3. Find the angle θ between two vectors a and b with magnitudes $\sqrt{3}$ and 2 respectively, having $\mathbf{a} \cdot \mathbf{b} = \sqrt{6}$:

(a) $\pi/6$ (b) $\pi/4$

(c) $\pi/3$ (d) $\pi/2$

Q4. If a is a unit vector and $(\mathbf{x} - \mathbf{a}) \cdot (\mathbf{x} + \mathbf{a}) = 15$, then find the value of |x|:

(a) 3 (b) 4

(c) $\sqrt{15}$ (d) 16

Q5. Find the projection of the vector a on the vector b for the following cases:

(i) $\mathbf{a} = -2\mathbf{i} + 3\mathbf{j} - 5\mathbf{k}$ on $\mathbf{b} = \mathbf{i} + 4\mathbf{j} - \mathbf{k}$

(ii) $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ on $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$

(iii) $\mathbf{a} = \mathbf{i} + 3\mathbf{j} + 7\mathbf{k}$ on $\mathbf{b} = 7\mathbf{i} - \mathbf{j} + 8\mathbf{k}$

(iv) $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$ on $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$

Q6. Find a unit vector perpendicular to each of the vectors (a + b) and (a − b), where:

(i) $\mathbf{a} = 3\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$

(ii) $\mathbf{a} = 3\mathbf{i} - 3\mathbf{j} + 0\mathbf{k}$ and $\mathbf{b} = -\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$

(iii) $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$

Q7. Find the area of a parallelogram whose adjacent sides are given by the vectors $\mathbf{a} = 3\mathbf{i} + \mathbf{j} + 4\mathbf{k}$ and $\mathbf{b} = \mathbf{i} - \mathbf{j} + \mathbf{k}$.

UNIT 2: THREE DIMENSIONAL GEOMETRY

Q8. Given the Cartesian equation of line: $(x - 2) / 2 = (2y - 5) / (-3) = z + 1$. If its vector equation is given by $\mathbf{r} = (2\mathbf{i} + (5/2)\mathbf{j} - \mathbf{k}) + \lambda(2\mathbf{i} - (3/2)\mathbf{j} + p\mathbf{k})$, then determine the value of p:

(a) 0 (b) 1

(c) 2 (d) 3

Q9. Find the direction cosines of the line passing through the two points:

(i) Point A(−2, 4, −5) and Point B(1, 2, 3)

(ii) Point A(6, 4, −5) and Point B(−2, 2, 3)

(iii) Point A(4, 3, −5) and Point B(−2, 1, −8)

(iv) Point A(2, 3, 5) and Point B(−1, 3, 2)

Q10. Find the shortest distance between the pair of lines given below:

(i) $\mathbf{r} = (3\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}) + \lambda(\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ and $\mathbf{r} = (-\mathbf{i} - \mathbf{j} - \mathbf{k}) + \mu(7\mathbf{i} - 6\mathbf{j} + \mathbf{k})$

(ii) $\mathbf{r} = (3\mathbf{i} - 5\mathbf{j} + \mathbf{k}) + \lambda(\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ and $\mathbf{r} = (-\mathbf{i} - \mathbf{j} + \mathbf{k}) + \mu(7\mathbf{i} - 6\mathbf{j} + \mathbf{k})$

(iii) $(x + 1)/7 = (y + 1)/(-6) = (z + 1)/1$ and $(x - 3)/1 = (y - 5)/(-2) = (z - 7)/1$

(iv) $\mathbf{r} = (\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) + \lambda(\mathbf{i} - 3\mathbf{j} + 2\mathbf{k})$ and $\mathbf{r} = (4\mathbf{i} + 5\mathbf{j} + 6\mathbf{k}) + \mu(2\mathbf{i} + 3\mathbf{j} + \mathbf{k})$

UNIT 3: LINEAR PROGRAMMING (LPP)

Q11. The corner points of the feasible region for an LPP are P(4, 3), Q(3, 6), R(2, 2) and S(12, 1). The maximum value of $Z = 5x - 2y$ occurs at:

(a) P (b) Q

(c) R (d) S

Q12. The corner points of the feasible region for an LPP are P(0, 5), Q(1, 5), R(4, 2) and S(12, 0). The minimum value of $Z = 2x + 5y$ occurs at:

(a) P (b) Q

(c) R (d) S

Q13. Solve the following Linear Programming Problems graphically:

(i) Maximise $Z = 3x + 4y$ subject to constraints: $x + y \leq 4$, $x \geq 0$, $y \geq 0$.

(ii) Maximise $Z = 2x + 6y$ subject to constraints: $x + y \leq 4$, $x \geq 0$, $y \geq 0$.

(iii) Minimise $Z = 200x + 500y$ subject to constraints: $x + 2y \geq 10$, $3x + 4y \leq 24$, $x \geq 0$, $y \geq 0$.

(iv) Maximise & Minimise $Z = 5x + 10y$ subject to constraints: $x + 2y \leq 120$, $x + y \geq 60$, $x - 2y \geq 0$, $x \geq 0$, $y \geq 0$.

UNIT 4: PROBABILITY

Q14. If $P(A) = 4/5$ and $P(A \cap B) = 7/10$, then the conditional probability $P(B/A)$ is equal to:

(a) 1/8 (b) 17/20

(c) 7/8 (d) 1/10

Q15. Evaluation of Independent Events:

(i) If A and B are two independent events such that P(A) = 0.4 and P(B) = 0.5, find P(B/A).

(ii) If A and B are two independent events such that P(A) = 0.3 and P(B) = 0.4, find P(B/A).

(iii) If P(A) = 3/5 and P(B) = 1/5, find P(A ∩ B) if A and B are independent events.

(iv) Given that the events A and B are such that P(A) = 1/2, P(A ∪ B) = 3/5 and P(B) = p. Find p if they are independent.

Q16. CASE STUDY PROBLEM 1: Flower Seed Germination

A shopkeeper sells three types of flower seeds A_1, A_2, and A_3. They are packaged as a mixture where the proportions of these seeds are 4 : 3 : 2 respectively. Based on historical data, the germination probabilities for types A_1, A_2, and A_3 are 45%, 60%, and 35% respectively.

(i) Calculate the probability that a randomly chosen seed will germinate.

(ii) Calculate the probability that the seed is of type A_1 given that a randomly chosen seed germinates.

— OR —

(ii) Calculate the probability that the seed is of type A_3, given that a randomly chosen seed germinates.

Q17. CASE STUDY PROBLEM 2: Seed Mixture Analysis

A agricultural retailer sells three types of flower seeds A, B, and C as a mixture combined in proportions 4 : 4 : 2. The germination rates for seeds A, B, and C are measured at 45%, 60%, and 35% respectively.

(i) Calculate the total probability of germination for any randomly selected seed.

(ii) Calculate the probability that the seed is of type B, given that it successfully germinates.

— OR —

(ii) Calculate the probability that the seed is of type A, given that it successfully germinates.

Q18. CASE STUDY PROBLEM 3: Multiple Choice Assessment Analysis

In answering a question on a multiple choice test, a student either knows the answer or guesses. Let 3/4 be the probability that he knows the answer and 1/4 be the probability that he guesses. Assuming that a student who guesses at the answer will be correct with probability 1/4.

(i) What is the probability that the student knows the answer given that he answered it correctly?